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CSC 345 Analysis of Discrete Structures

Class meetings	Spring 2025 Tuesdays and Thursdays, 3:30-4:45pm in-person at CESL 103 via Zoom at arizona.zoom.us/j/89848283171
Dates	January 16 First class March 11 and 13 Spring recess May 6 Last class
Course description	This is a course on <i>data structures</i> and their fundamental <i>algorithms</i>. Data structures (a collection of <i>data</i>, together with a repertoire of <i>operations</i> that can be efficiently performed on the data) are the building blocks of algorithms. We will study the classic data structures of <i>lists</i>, <i>trees</i>, <i>tables</i>, <i>heaps</i>, and <i>graphs</i>. The course begins with an introduction to algorithms and their analysis, applies this to the fundamental problem of <i>sorting</i>, and then studies the classic data structures and their efficient implementation. We conclude with an introduction to elementary graph algorithms. The emphasis is on learning how to <i>design</i>, <i>analyze</i>, and <i>implement</i> data structures. These skills are developed through written assignments containing challenging exercises, and programming projects that make these data structures come alive.
Official catalog description	The official catalog description of the course is available at the University course catalog here .
Prerequisites	CSC 210 and CSC 244
Instructor	John Kececioglu kece@cs.arizona.edu Gould-Simpson 727 (520) 621-4526
Assistants	Hamad Ayaz (undergraduate) hamadayaz@arizona.edu Fay Garcia (undergraduate) faygarcia@arizona.edu Asiful Islam (graduate) asifulislam@arizona.edu Hyungji Kim (undergraduate) hyungjikim@arizona.edu Andres Silva-Castellano (undergraduate) andressc@arizona.edu Moyeen Uddin (graduate) moyeen@arizona.edu

Office hours	Mondays	2-3pm	Hamad	Gould-Simpson 934 (TA)
	Tuesdays	10-11am	John	arizona.zoom.us/j/5128057331 (Professor)
	Tuesdays	11am-12pm	Fay	Gould-Simpson 934 (TA)
	Wednesdays	11am-12pm	Asiful	Gould-Simpson 934 (TA)
	Wednesdays	1:30-2:30pm	John	Gould-Simpson 727 (Professor)
	Thursdays	12-1pm	Andres	Gould-Simpson 934 (TA)
	Thursdays	1-2pm	Asiful	Gould-Simpson 934 (TA)
	Fridays	12-1pm	Hyungji	Gould-Simpson 934 (TA)
	Fridays	2:30-3:30pm	Moyeen	Gould-Simpson 934 (TA)
Web	Course homepage	www.cs.arizona.edu/classes/cs345/spring25/		
	Instructor homepage	www.cs.arizona.edu/people/kece/		
	D2L site	d2l.arizona.edu/d2l/home/1553361		
Course format	The course is taught <i>in-person</i> in a lecture format, along with a simultaneous Zoom <i>stream</i> whose video is recorded. Lectures are given by the instructor. Homeworks, exams, and programming projects all involve <i>individual</i> work.			
Course objectives	The teaching objectives for the course are basic <i>algorithm analysis</i> techniques; classic <i>sorting algorithms</i> ; the fundamental <i>data structures</i> of linked lists, hash tables, priority queues, and balanced search trees; and elementary <i>graph algorithms</i> .			
Learning outcomes	Students successfully completing the course will <i>learn</i> the fundamental <i>sorting algorithms</i> and <i>data structures</i> that are widely used throughout computer science, and will gain the capacity to <i>implement</i> basic <i>data structures</i> . Skill in algorithms will be acquired through problem solving on homeworks and exams, while skill in implementation will be acquired through programming projects.			
Transferable career skills	(National Association of Colleges and Employers Career Readiness) Career readiness is a foundation from which to demonstrate requisite <i>core competencies</i> that broadly prepare the college-educated for success in the workplace and lifelong career management. For new college graduates, career readiness is key to ensuring successful entrance into the workforce.			
	In this course, we focus on the following core competencies. • <i>Critical thinking</i>: Students will learn to design and analyze data structures. • <i>Communication</i>: Students will learn to communicate data structures and their analysis to others through written assignments. • <i>Technology</i>: Students will develop and test programs that implement data structures.			
Course communications	Online communication outside the lectures will be via: postings in the announcements section at the bottom of this webpage, or through announcements on the D2L site for the course, and by email with the instructor at the address given above .			
Required text	Thomas H. Cormen, Charles E. Leiserson, Ron L. Rivest, and Clifford Stein, Introduction to Algorithms , fourth edition, MIT Press, Cambridge, Massachusetts, 2022.			
Related books	The following are also excellent books on data structures. Harry R. Lewis and Larry Denenberg, Data Structures and Their Algorithms , Addison-Wesley, New York, 1991.			

Kurt Mehlhorn and Peter Sanders,
[*Algorithms and Data Structures: The Basic Toolbox*](#),
Springer, Berlin, 2008.

Homeworks

There are six *homeworks*, which occur roughly every two weeks.

- [Homework 1](#), due February 6 at 3:30pm ([solutions](#)).
- [Homework 2](#), due February 20 at 3:30pm ([solutions](#)).
- [Homework 3](#), due February 27 at 3:30pm ([solutions](#)).
- [Homework 4](#), due March 27 at 3:30pm ([solutions](#)).
- [Homework 5](#), due April 10 at 3:30pm ([solutions](#)).
- [Homework 6](#), due April 25 at 5:00pm ([solutions](#)).

Of these six homeworks, the one having the *lowest score* will be dropped. So your overall homework grade will be based on your *five best* homework scores.

Exams

There are two *exams*: a midterm, and a final.

- Midterm exam, March 6.
- Final exam, May 14, 3:30-5:30pm.

The *final* is comprehensive. The University's final exam regulations are [here](#), and the University's final exam schedule is [here](#).

Projects

There are two *programming projects*.

- [Project 1](#), due April 3 at 11:59pm.
- [Project 2](#), due May 1 at 11:59pm.

Grading

Course grades use the regular scale of A, B, C, D, and E. Grades are determined from a weighted average of the percentage of points earned on homeworks, programming projects, and exams.

25%	Homeworks
20%	Programming projects
25%	Midterm exam
30%	Final exam

Attaining a weighted score of at least 90% guarantees an A in the course, at least 80% guarantees a B, at least 70% guarantees a C, and at least 60% guarantees a D. These *thresholds* may be lowered, but will not be raised.

The awarding of *points* on homework and exam questions is roughly according to the following scheme. Having the right solution *idea* and the right technical *execution* of this idea earns at least 90%. (Only a perfect write-up that cannot be improved earns 100%.) Having the right idea but with errors in its execution is at least 80%. Having the wrong idea and errors in its execution, but demonstrating comprehension of the material, is at least 70%. Having the wrong idea, errors in execution, and deficiencies in comprehension, is roughly 60%. Work that shows no understanding is roughly 50%.

On homework and exam questions, writing an answer that relates to the question *guarantees* at least 50% of the points for the question (which is a failing percentage). No points are awarded for writing nothing.

Homeworks may contain *bonus problems*. Points earned on bonus questions are not added to points for required questions. Bonus points are tallied separately at the end of

the semester, and considered subjectively as a measure of effort when deciding whether to move up a student who is near a letter-grade boundary.

On *homework*, very-high-level ideas can be discussed with friends, but solutions must represent individual work and must be written up separately. Use of solutions from previous offerings of the course is not permitted. Any material from the Internet that is used in a solution must be cited by its URL; to not cite it is *plagiarism*, which is considered cheating. (For more information, see the Department of Computer Science [Course Policy on Collaboration](#) and the University [Code of Academic Integrity](#).)

When turning in *solutions* to homeworks, start each problem on a separate page, and put the problems in the correct order. Neatness, and especially conciseness, is required to earn the highest marks. If a problem eludes solution, state this up front and write down only what you know to be correct. Rambling at length about attempts that didn't succeed may result in more points being deducted.

Homework that is *late* (submitted after the due date and time) is not accepted.

Incomplete or withdrawal Requests for an *incomplete* (I) or *withdrawal* (W) must be made in accordance with University policies, which are available [here](#).

Regrades Requests for *regrades* of homeworks or exams must be submitted within one week of receiving the graded homework or exam.

Attendance

While attendance will not be taken or recorded, students are expected to *attend* all lectures to keep up with the course content (which builds in successive lectures across the semester). When a student must miss a lecture, they should obtain notes of the missed lecture from a classmate. (Watching the recorded video of the Zoom stream for the missed lecture may be helpful, but is not intended to substitute for attending in-person.)

Honors credit

Students who wish to take the course for *honors credit* should [email](#) the [instructor](#) to make an appointment to discuss the terms and to sign the honors course request [form](#).

Course topics

We cover the topics of the course in seven *parts*. After each topic below is the appropriate *chapter* number from the text.

I Introduction

- *Algorithms* (1, 2): notion of algorithm, measuring time and space.
- *Analysis* (3, 4): asymptotics, sums, recurrences.

II Sorting

- *Merge sort* (2): sorting by divide and conquer.
- *Heap sort* (6): sorting using the implicit heap data structure.
- *Quick sort* (optional) (7): sorting efficiently with respect to expected time.
- *Counting sort* (8): sorting integers in a bounded range.
- *Radix sort* (8): sorting tuples in lexicographic order.

III Elementary data structures

- *Lists* (10): endogenous and exogenous lists; doubly-linked, circular, anchored lists.
- *Trees* (10): binary and multiway trees; tree traversals.

IV Tables

- *Hashing with buckets* (11): hash tables, hash functions, chaining collisions.
- *Perfect hashing* (optional) (11): eliminating collisions.

V Search trees

- *Basic operations* (12): search, insertion, deletion, predecessor, and successor.
- *Red-black trees* (13): worst-case efficiency through height balancing.
- *Splay trees* (optional) (outside material): amortized efficiency through self-organizing splaying.
- *Augmenting search trees* (17): attributing a data structure to support new operations.

VI Heaps

- *Implicit heaps* (6): array-based heap in heap sort.

VII Graphs

- *Representation* (20): basic concepts, adjacency matrix, adjacency list.
- *Searching* (20): depth-first and breadth-first search of undirected and directed graphs, connected components.
- *Topological sort* (optional) (20): ordering directed acyclic graphs.
- *Strongly-connected components* (optional) (20): decomposing into components having bidirectional paths.

Weekly schedule

We cover these topics in the following *weekly schedule*. (This schedule is tentative, and subject to adjustment based on instructor discretion. When appropriate, relevant readings in the sections of the textbook, and links to handouts, will be posted.)

	January 16	Course overview
January 21	January 23 (Homework 1 out)	Algorithm analysis; insertion sort
January 28	January 30	Algorithm analysis; merge sort
February 4	February 6 (Homework 1 due) (Homework 2 out)	Heap sort
February 11	February 13 (Homework 1 back)	Counting sort
February 18	February 20 (Homework 2 due) (Homework 3 out)	Radix sort
February 25 (Homework 2 back)	February 27 (Homework 3 due)	Quick sort

March 4 (Homework 3 back)	March 6 (Midterm exam)	Exam review; midterm exam
March 11 (No class)	March 13 (No class)	(Spring break week)
March 18 (Homework 4 out)	March 20 (Midterm exam back)	Lists, stacks, queues, and dequeues
March 25 (Project 1 out)	March 27 (Homework 4 due) (Homework 5 out)	Trees; hashing with chaining
April 1	April 3 (Homework 4 back) (Project 1 due)	Basic search trees
April 8	April 10 (Homework 5 due) (Homework 6 out) (Project 1 back)	Red-black trees
April 15 (Project 2 out)	April 17 (Homework 5 back)	Red-black trees
April 22	April 24 (Homework 6 due)	Augmenting data structures
April 29	May 1 (Homework 6 back) (Project 2 due)	Augmenting data structures
May 6 (Project 2 back)		Final exam review

Classroom behavior

To foster a *positive learning environment*, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (such as texting, chatting, reading a newspaper, making phone calls, web surfing, and so on).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in *disruptive activity* will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

Some learning styles are best served by using personal electronics, such as laptops and iPads. These devices can be distracting to other learners. Therefore, students who prefer to use *electronic devices* for note-taking during lecture should use one side of the classroom.

Safety on campus

For a list of *emergency procedures* for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT), available [here](#).

Please also watch the video available [here](#).

University-wide policies

Links to the following *University policies* (listed below) are available [here](#).

- Absence and class participation policies
- Threatening behavior policy
- Accessibility and accommodations policy
- Code of academic integrity
- Nondiscrimination and anti-harrassment policy

Department-wide policies

Links to the following *Department policies* (listed below) are available [here](#).

- Department code of conduct
- Class recordings
- Illnesses and emergencies
- Obtaining help
- Preferred names and pronouns
- Confidentiality of student records
- Additional resources

Land Acknowledgement Statement

We respectfully acknowledge the University of Arizona is on the *land and territories of indigenous peoples*. Today, Arizona is home to twenty-two federally recognized tribes, with Tucson being home to the O’odham and the Yaqui. Committed to diversity and inclusion, the University strives to build sustainable relationships with sovereign native nations and indigenous communities through education offerings, partnerships, and community service.

Subject to change

Information contained in the course syllabus, other than the grade and absence policy, may be *subject to change* with advance notice, as deemed appropriate by the instructor.

Distributions

Grade *distributions* are available [here](#).

Announcements

The *grades* for the Final Exam have been published on [Gradescope](#).

The *distribution* of grades for the Final Exam, and Overall in the course, are available [above](#).

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<https://www.cs.arizona.edu/classes/cs345/spring25/>

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18 May 2025